

THENMOZHI SEKAR

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EDUCATION

Quinnipiac University,

Master of Science in Computer Science, **GPA – 4.0**

Hamden, CT, USA

Aug 2024 – May 2026

Madras University,

Bachelor of Computer Applications, **GPA - 3.7**

Kanchipuram, TN, India

Aug 2018 – May 2021

EXPERIENCE

Graduate Research Assistant (AI/Software Engineering)

Aug 2024 – Present

Quinnipiac University,

Hamden, CT, USA

- Conducted weekly literature reviews in **AI/ Software Engineering** to identify research gaps and emerging methodologies.
- Collaborated with two faculty advisors to develop and refine research ideas, drafting project proposals, and transforming concepts into publishable research topics.
- Designed and implemented **data engineering and machine learning pipelines**, including data collection, cleaning, using multiple LLMs for AI Research, and experimental analysis using Python and Java, and contributed to a **research publication**

Full Stack Developer

Jun 2022 – April 2024

Goaira Technologies Pvt Limited,

Chennai, India

- Analyzed system performance and user issue data to identify patterns, improving reliability across **200+ devices**
- Automated monitoring and maintenance workflows using scripting, reducing manual effort by **30%** and improving efficiency
- Collected and documented technical issue data to support **data-driven decision-making** and enable predictive maintenance insights.
- Collaborated with cross-functional teams to align solutions with production requirements, improving deployment efficiency

Full Stack Java Developer

Nov 2021 – May 2022

Ijona Technologies Pvt. Ltd,

Bangalore, India

- Developed and tested **algorithmic models** using experimental analysis to improve performance and results
- Analyzed **user behavior data** to optimize system design and enhance overall performance
- Improved system reliability and efficiency through testing, performance analysis, and continuous optimization

PROJECTS

AI-Based Code Generation Evaluation | Python, Java, Published Research

Aug 2025 – Feb 2026

- Conducted empirical evaluation of AI-generated code on **200+ code samples**, performing **1200+ experimental runs** using similarity metrics and structural analysis
- Compared AI-generated code with human-written code using evaluation techniques- **sequence ratio** and **logical correctness**
- Published research paper at **IISEC 2026**, contributing to the field of AI in Software Engineering

LLM-Based Bug Report Enhancement Pipeline | Python, NLP, LLMs

Jan 2026 – Present

- Built an **LLM-based NLP pipeline** to convert unstructured bug reports into standardized formats, improving the report completeness by **72%**
- Processed **15,000+ real-world GitHub API bug reports**, extracting structured insights for analysis, triaging, and prediction tasks
- Enhanced debugging workflows by generating clear, consistent, and reproducible bug reports

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R

AI/SE/ML: Scikit-learn, XGBoost, Pandas, NumPy, PyTorch, NLP

Developer Tools: Git, Jira, Visual Studio, Docker, MLflow, OpenCV, AWS, Kubernetes, Airflow

Methodologies: Agile, SDLC, CI/CD, TDD

Concepts: Data Structures & Algorithms, OOP, APIs, System Design